Why is i^i a real number?

David Meyer

dmm613@gmail.com

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1 Introduction

There are a few ways to think about this question. But first we look at why a complex number z equals $re^{i\theta}$. Then in Section 3 we use the fact that complex conjugation is an automorphism of $\mathbb{C}$ to show that $i^i \in \mathbb{R}$. In Section 4 we use the power series expansion of i^i to show that $i^i \in \mathbb{R}$, and in Section 5 we rely on Euler's formula to show the same result. In Section 6 we do a bit of arithmetic to show the numerical value of i^i. Finally Section 7 offers a few conclusions.

Before launching into all of this, note that we will make heavy use of the exponential function $\exp(z)$ (and therefore as you might expect, this means heavy use of $\log(z)$, where by $\log(z)$ we mean $\log_e(z)$ aka $\ln(z)$). The important point for this discussion is that $\exp(z)$ is injective over $\mathbb{R}$ but not over $\mathbb{C}$. More specifically, for a complex number z the the complex logarithm $\log(z)$ is defined as the inverse function to the exponential function. That is, it satisfies $e^{\log z} \equiv z$. Now since $re^{i\theta} = re^{i\theta+2\pi ik}$ for all $k \in \mathbb{Z}$ we have that for any choice of k, $\log(re^{i\theta+2\pi ki})$ is a valid inverse for $re^{i\theta}$. The logarithm is therefore a multivalued function and each value of k defines what is called a different *branch* of the logarithm. The principal branch usually corresponds to the choice $k = 0$ [9].

Remark 1.1. Antoine Chambert-Loir (@antoinechambertloir@mathstodon.xyz) tells me that restricting the logarithm to any particular branch causes all algebraic relations break down. The example he gives is that $a^c \times b^c = (ab)^c$ only when the sum of the principal arguments of a and b equals the principal argument of ab [2]. Apparently a similar problem arises for conjugacy when a is strictly negative.

2 First: Why does a complex number z equal $r\mathrm{e}^{i\theta}$?

Our first question regards the nature of a complex number z, where the length of the line from the origin to the point z, $|z|$, equals r. In particular, why does a complex number $z = a + bi = r\mathrm{e}^{i\theta}$? Consider the complex plane, which is shown in Figure 1:

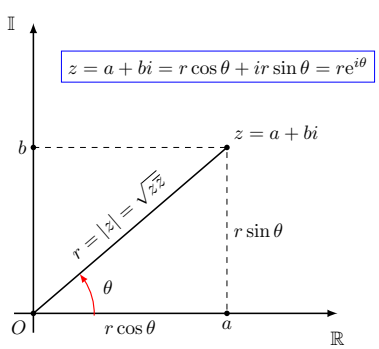


Figure 1: The Complex Plane

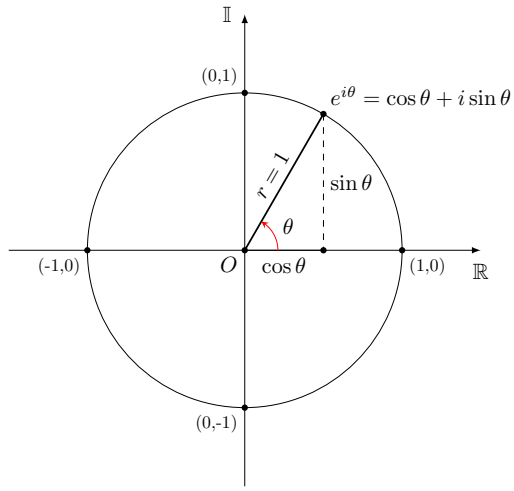


Figure 2: Euler's Formula, the Unit Circle, and the Complex Plane

Given Figures 1 and 2 it is pretty easy to see why $z = re^{i\theta}$:

$$\begin{aligned}
z &= a + ib && \text{\# definition of a point } z \text{ in the complex plane} \\
&= r\cos\theta + ir\sin\theta && \text{\# switch to polar coordinates: } a = r\cos\theta \text{ and } b = r\sin\theta \\
&= r\left(\cos\theta + i\sin\theta\right) && \text{\# factor out } r \\
&= re^{i\theta} && \text{\# } e^{i\theta} = \cos\theta + i\sin\theta \text{ (Euler's formula [8])}
\end{aligned}$$

So we see that $z = re^{i\theta}$. ■

3 Automorphism of $\mathbb{C}$

In this section we look at solving this puzzle using the fact that complex conjugation is an automorphism of $\mathbb{C}$ [4]. For this approach we need the following facts:

1. If a complex number z equals its complex conjugate $\overline{z}$ then $z \in \mathbb{R}$
2. $-i = i^{-1}$
3. $\overline{z} = re^{-i\theta}$

To see 1., consider the following argument: Let z be a complex number so that $z = a + bi$. Then $\overline{z} = a - bi$. So $z = \overline{z} \Rightarrow a + bi = a - bi$. Subtracting a from both sides of the right hand side of this implication gives us $bi = -bi$ or $2bi = 0$. Since we know that $2 \neq 0$ and $i \neq 0$ it must be the case that $b = 0$. Said another way: $\text{Im}(z) = 0$ and $\text{Re}(z) = z$. That is, $z \in \mathbb{R}$. So if $z = \overline{z}$ we know that $z \in \mathbb{R}$.

To see 2., notice that $-(i \cdot i) = -i^2 = -(-1) = 1$ and $-i \cdot i = 1 \Rightarrow -i = i^{-1}$.

To see 3., consider

$$\begin{aligned}
\overline{z} &= \overline{r \cdot (\cos\theta + i\sin\theta)} && \text{\# since } z = r(\cos\theta + i\sin\theta) \text{ (Figure 1)} \\
&= \overline{r} \cdot \overline{(\cos\theta + i\sin\theta)} && \text{\# since } \overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2} \text{ (product rule for complex conjugates [5])} \\
&= r \cdot \overline{(\cos\theta + i\sin\theta)} && \text{\# since } \overline{r} = \overline{r + 0i} = r - 0i = r \ (\overline{r} = r \text{ for } r \in \mathbb{R}) \\
&= r \cdot (\cos\theta - i\sin\theta) && \text{\# since } \overline{a + bi} = a - bi \\
&= re^{-i\theta} && \text{\# since } e^{-i\theta} = \cos\theta - i\sin\theta \text{ (corollary to Euler's formula [3])}
\end{aligned}$$

Now, we want to show that $\overline{i^i} = i^i$, which would show that $i^i \in \mathbb{R}$ by the above argument. So consider:

$$\begin{aligned}
\overline{i^i} &= \overline{i}^{\overline{i}} && \text{\# see proof below} \\
&= (-i)^{-i} && \text{\# since } i = 0 + 1i \text{ and } \overline{0 + 1i} = -1i = -i \\
&= \left(i^{-1}\right)^{-i} && \text{\# since } -i = i^{-1} \text{ (see item 2. above)} \\
&= i^{(-1 \cdot -i)} && \text{\# since } (x^m)^n = x^{mn} \\
&= i^i && \text{\# since } -1 \cdot -i = i
\end{aligned}$$

So we see that $\overline{i^i} = i^i$ and so $i^i \in \mathbb{R}$. ■

All ok, but why does $\overline{i^i} = \overline{i}^{\overline{i}}$? To see why first notice that

$$\begin{aligned}
\log(\overline{z}) &= \log(\overline{re^{i\theta}}) && \text{\# since } z = re^{i\theta} \text{ (Figure 1)} \\
&= \log(re^{-i\theta}) && \text{\# by item 3. above} \\
&= \log\left(\frac{r}{e^{i\theta}}\right) && \text{\# since } a^{-b} = \frac{1}{a^b} \\
&= \log(r) - \log(e^{i\theta}) && \text{\# by the quotient rule for logarithms [7]}
\end{aligned}$$

So

$$\log(\overline{z}) = \log(r) - \log(e^{i\theta}) \tag{1}$$

Next, notice that

$$\begin{aligned}
\overline{\log(z)} &= \overline{\log(re^{i\theta})} && \text{\# since } z = re^{i\theta} \\
&= \overline{\log(r) + \log(e^{i\theta})} && \text{\# by the product rule for logarithms} \\
&= \overline{\log(r)} + \overline{\log(e^{i\theta})} && \text{\# by the sum rule for conjugates [6]} \\
&= \log(r) + \overline{\log(e^{i\theta})} && \text{\# since } \log(r) \in \mathbb{R} \\
&= \log(r) + \overline{i\theta} && \text{\# since } \log(e^{i\theta}) = i\theta \\
&= \log(r) - i\theta && \text{\# since } \overline{i\theta} = \overline{0 + i\theta} = -i\theta \\
&= \log(r) - \log(e^{i\theta}) && \text{\# since } i\theta = \log(e^{i\theta}) \\
&= \log(\overline{z}) && \text{\# by Equation (1)}
\end{aligned}$$

Now we know that

$$\log(\overline{z}) = \overline{\log(z)} \tag{2}$$

and so we can see that

$$\begin{aligned}
\log(\overline{z^w}) &= \overline{\log(z^w)} && \text{\# Equation (2)} \\
&= \overline{w \log(z)} && \text{\# by the power rule for logarithms} \\
&= \overline{w} \cdot \overline{\log(z)} && \text{\# by the product rule for conjugation} \\
&= \overline{w} \cdot \log(\overline{z}) && \text{\# by Equation (2)} \\
&= \log(\overline{z}^{\overline{w}}) && \text{\# by the power rule for logarithms}
\end{aligned}$$

So now we know that

$$\log(\overline{z^w}) = \log(\overline{z}^{\overline{w}}) \tag{3}$$

Exponentiating both sides of Equation (3) gives us

$$\overline{z^w} = \overline{z}^{\overline{w}} \tag{4}$$

So now if we set $z = w = i$ in Equation (4) we see that $\overline{i^i} = \overline{i}^{\overline{i}}$. ■

Note that this argument relies on the fact that complex conjugation is an automorphism of $\mathbb{C}$. So one "problem" with this approach is that you would first need to know not only that conjugation is an automorphism of $\mathbb{C}$ but also that the automorphism is equipped with exponentiation. So one could argue that this is proof is incomplete since it may not be true for branches (of the logarithm) other than the principal branch [9].

4 Power Series

We know that the Taylor series [10] for $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$. We also know from Euler's formula [1] that $i = e^{i\frac{\pi}{2}}$ (since $e^{i\frac{\pi}{2}} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i$). Note that the same result can be obtained with $x \in \left\{\frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \ldots\right\}$, so in general $x = \frac{\pi}{2} \cdot (1 + 4n)$ where $n \in \mathbb{N} \cup \{0\}$.

So we see that

$$\begin{aligned}
i^t &= \left(e^{\frac{i\pi}{2}}\right)^t && \text{\# Euler's formula for } i\text{, raised to the power } t \\
&= e^{\frac{it\pi}{2}} && \text{\# since } (x^m)^n = x^{mn} \\
&= \sum_{k=0}^{\infty} \frac{\left(\frac{it\pi}{2}\right)^k}{k!} && \text{\# since } e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \text{ and } x = \frac{it\pi}{2}
\end{aligned}$$

If we let $t = i$ we get $i^i = e^{i^2\frac{\pi}{2}} = e^{-\frac{\pi}{2}}$. So $i^i = e^{-\frac{\pi}{2}}$ and since $e^{-\frac{\pi}{2}} \in \mathbb{R}$ we know that $i^i \in \mathbb{R}$. Fortunately this agrees with the result we find in Section 5. ■

5 Euler's Formula

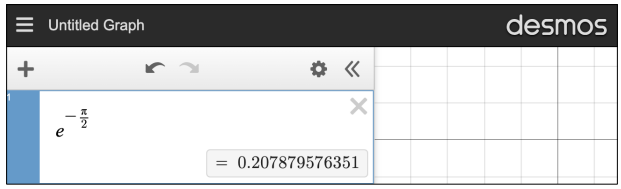
Another approach to answering this question is to consider Euler's formula evaluated at $x = \frac{\pi}{2}$:

$$\begin{aligned}
e^{ix} &= \cos x + i\sin x && \text{\# Euler's formula} \\
&\Rightarrow e^{i\frac{\pi}{2}} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} && \text{\# set } x = \frac{\pi}{2} \\
&\Rightarrow e^{i\frac{\pi}{2}} = 0 + i \cdot 1 && \text{\# } \cos\frac{\pi}{2} = 0 \text{ and} \sin\frac{\pi}{2} = 1 \\
&\Rightarrow e^{i\frac{\pi}{2}} = i && \text{\# simplify} \\
&\Rightarrow \left(e^{i\frac{\pi}{2}}\right)^i = i^i && \text{\# raise both sides to } i \\
&\Rightarrow e^{\frac{i^2\pi}{2}} = i^i && \text{\# } (x^m)^n = x^{mn} \\
&\Rightarrow e^{-\frac{\pi}{2}} = i^i && \text{\# } i^2 = -1 \\
&\Rightarrow e^{-\frac{\pi}{2}} \in \mathbb{R} \Rightarrow i^i \in \mathbb{R} && \text{\# } i^i \text{ is a real number}
\end{aligned}$$

It is important to notice that this expression is multivalued, and when we evaluate it at $\frac{\pi}{2}$ we are on the principle branch. However, the same result can be obtained with $x \in \left\{\frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \ldots\right\}$. In general $x = \frac{\pi}{2} \cdot (1 + 4n)$ for $n \in \mathbb{N} \cup \{0\}$. ■

6 Ok, $i^i \in \mathbb{R}$, but what does it equal

We saw above that $i^i = e^{-\frac{\pi}{2}}$. So $i^i = e^{-\frac{\pi}{2}} \approx 0.20788$.



7 Conclusions

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LaTeX Source

`https://www.overleaf.com/read/cfjttjzdbfnc`

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